

Topic 07: Special Functions

Unit IV — Laplace Transforms

JNTUA College of Engineering (Autonomous) ATP

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Hook

A light switch turns on at $t = 2$ seconds. An explosion happens instantaneously at $t = 3$ seconds. A filter processes a signal with a fractional power decay $t^{-1/2}$. None of these can be described using elementary functions alone — you need special functions. The unit step function captures the switch, the Dirac delta captures the explosion, and the Gamma function captures the fractional power. Each has a Laplace transform. This topic gives you all three, plus two more.

1 Why This Topic Exists

Before we begin, here is the honest answer to why you are reading this...

Real engineering signals include sudden switches, impulse forces, and fractional powers — none of which are handled by the elementary functions covered in Topics 04–06. If you try to model a circuit that switches on at $t = 2$ s with a sine or exponential, you will obtain the wrong differential equation. The special functions in this topic were invented precisely to fill these gaps.

Mistake	Consequence
Not knowing the unit step function	Unable to represent piecewise inputs compactly; forced into case-by-case analysis that breaks when solving ODEs systematically.
Treating $\delta(t)$ as an ordinary function with a finite value	Mathematical errors in impulse response calculations; integrals of δ give wrong answers.
Ignoring the Gamma function	Cannot compute $\mathcal{L}\{t^n\}$ for non-integer n ; entire class of filter and signal problems becomes inaccessible.

Key Takeaway

Every special function in this topic was invented to solve a real problem. Treat each one as a tool with a specific job, and your toolkit becomes significantly more powerful.

2 You Already Know This (Intuition First)

Think of a craftsman's toolbox. When you first learn woodworking you buy a hammer, a screwdriver, and a saw — the elementary tools. These handle the vast majority of jobs. But the moment you need to tighten a bicycle wheel you reach for a torque wrench; when wiring a circuit you grab wire strippers. These are *specialised* tools, invented because a specific job kept coming up and none of the general tools handled it cleanly.

Special functions work the same way. The elementary functions — powers, exponentials, sines — are the hammer and screwdriver. The unit step function, Dirac delta, Gamma function, error function, and Bessel functions are the specialised tools. Each was

invented or formalised because a specific class of engineering problem appeared repeatedly and the basic toolkit fell short.

By the end of this topic you will have five new tools in your transform toolkit, each with a clear Laplace transform and a clear use-case.

3 The Unit Step Function (Heaviside Function)

Definition and Laplace Transform of $u(t - a)$

Definition. The *unit step function* (or Heaviside function) is

$$u(t - a) = \begin{cases} 0 & t < a \\ 1 & t \geq a \end{cases}$$

also written $H(t - a)$. The special case $a = 0$ gives $u(t) = u(t - 0)$, which is 1 for all $t \geq 0$ and 0 for $t < 0$.

Laplace Transform — Derivation.

$$\begin{aligned} \mathcal{L}\{u(t - a)\} &= \int_0^{\infty} e^{-st} u(t - a) dt \\ &= \int_a^{\infty} e^{-st} dt \quad (\text{since } u(t - a) = 0 \text{ for } t < a) \\ &= \left[\frac{e^{-st}}{-s} \right]_a^{\infty} \\ &= 0 - \frac{e^{-as}}{-s} \\ &= \boxed{\frac{e^{-as}}{s}}, \quad s > 0. \end{aligned}$$

Writing Piecewise Functions Using Unit Steps. Any piecewise function

$$f(t) = \begin{cases} g_1(t) & 0 \leq t < a \\ g_2(t) & t \geq a \end{cases}$$

can be written as

$$f(t) = g_1(t) + [g_2(t) - g_1(t)] u(t - a).$$

This converts a case-by-case definition into a single algebraic expression that can be transformed term-by-term using the Second Shifting Theorem:

$$\mathcal{L}\{g(t - a) u(t - a)\} = e^{-as} G(s), \quad G(s) = \mathcal{L}\{g(t)\}.$$

3.1 Worked Example: Piecewise Function via Unit Steps

Example — Writing a Piecewise Function and Finding Its Transform

Problem. Let

$$f(t) = \begin{cases} 3t & 0 \leq t < 4 \\ 12 & t \geq 4 \end{cases}.$$

Express $f(t)$ using unit step functions and hence find $\mathcal{L}\{f(t)\}$.

Step 1 — Rewrite using unit steps.

Here $g_1(t) = 3t$ and $g_2(t) = 12$.

$$f(t) = 3t + (12 - 3t)u(t - 4).$$

Note that $12 - 3t = -3(t - 4) + 0$, so we write $12 - 3t = -3(t - 4)$ evaluated at general t . We need to express $(12 - 3t)$ in terms of $(t - 4)$ for the shifting theorem:

$$12 - 3t = -3(t - 4).$$

Hence

$$f(t) = 3t - 3(t - 4)u(t - 4).$$

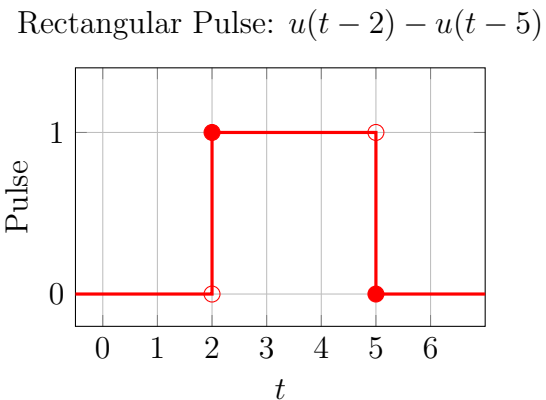
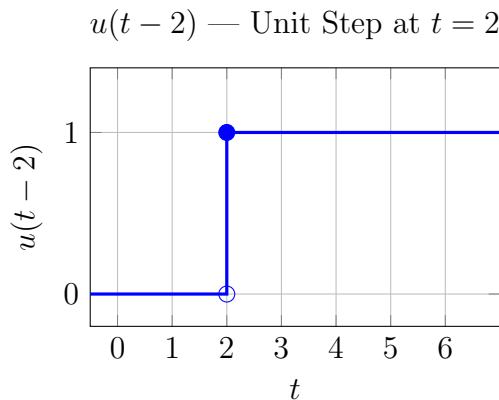
Step 2 — Apply the Laplace transform.

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \mathcal{L}\{3t\} - 3\mathcal{L}\{(t - 4)u(t - 4)\} \\ &= 3 \cdot \frac{1}{s^2} - 3e^{-4s} \cdot \frac{1}{s^2} \quad (\text{Second Shifting Theorem with } g(t) = t) \\ &= \frac{3}{s^2} - \frac{3e^{-4s}}{s^2} \\ &= \boxed{\frac{3(1 - e^{-4s})}{s^2}}. \end{aligned}$$

What Did We Learn?

To use the Second Shifting Theorem, rewrite every piece of the piecewise function so that each factor alongside $u(t - a)$ is expressed as a function of $(t - a)$, not bare t . Then $\mathcal{L}\{g(t - a)u(t - a)\} = e^{-as}G(s)$ applies directly.

3.2 Graphs of the Unit Step Function



Reading the Graphs

$u(t - 2)$ stays zero until $t = 2$, then jumps to 1 and stays there. The rectangular pulse $u(t - 2) - u(t - 5)$ is 1 only on the interval $[2, 5)$ and zero everywhere else — the standard way to “window” a signal.

4 The Dirac Delta Function (Impulse Function)

Definition and Laplace Transform of $\delta(t - a)$

Motivation. Think of a hammer blow on a structure. The force is enormous but lasts only an instant. Mathematically, model it as the limit of a rectangular pulse of height $1/\varepsilon$ and width ε as $\varepsilon \rightarrow 0$. The area (impulse) is always 1.

Defining properties:

$$\delta(t - a) = 0 \quad \text{for } t \neq a, \quad \int_{-\infty}^{\infty} \delta(t - a) dt = 1.$$

Sifting property. For any function f continuous at $t = a$:

$$\int_{-\infty}^{\infty} f(t) \delta(t - a) dt = f(a).$$

This is the defining operational property of $\delta(t - a)$.

Laplace Transform — Derivation using the sifting property.

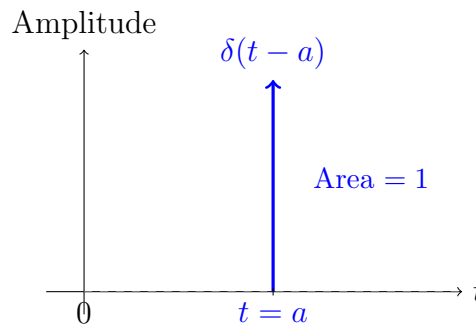
$$\begin{aligned} \mathcal{L}\{\delta(t - a)\} &= \int_0^{\infty} e^{-st} \delta(t - a) dt \\ &= e^{-sa} \quad (\text{sifting property with } f(t) = e^{-st}, a \geq 0) \\ &= \boxed{e^{-as}}, \quad a \geq 0. \end{aligned}$$

Special case $a = 0$: $\mathcal{L}\{\delta(t)\} = 1$.

Engineering applications:

- **Impulsive mechanical force:** a hammer blow at time a modelled as $F_0 \delta(t - a)$.
- **Instantaneous charge injection:** in a circuit, injecting charge Q at $t = a$ gives a current $I(t) = Q \delta(t - a)$.
- **Initial condition modelling:** a system at rest but with a non-zero initial velocity can be modelled using $\delta(t)$ forcing.

4.1 Schematic Diagram of the Dirac Delta



What Did We Learn?

The Dirac delta $\delta(t - a)$ is not a function in the classical sense — it is a *distribution*. Its key feature is the sifting property: it samples the value of any continuous function at the single point $t = a$. Its Laplace transform e^{-as} is simpler than the unit step transform e^{-as}/s because there is no integration of the constant 1 over time.

5 The Gamma Function

Definition

Definition.

$$\Gamma(n) = \int_0^\infty t^{n-1} e^{-t} dt, \quad n > 0.$$

Reduction formula (proof by integration by parts).

Let $u = t^{n-1}$, $dv = e^{-t} dt$. Then $du = (n-1)t^{n-2} dt$, $v = -e^{-t}$.

$$\begin{aligned} \Gamma(n) &= \left[-t^{n-1} e^{-t} \right]_0^\infty + (n-1) \int_0^\infty t^{n-2} e^{-t} dt \\ &= 0 + (n-1) \Gamma(n-1). \end{aligned}$$

Therefore

$$\Gamma(n+1) = n \Gamma(n) \quad (\text{reduction formula}).$$

Values for positive integers:

- $\Gamma(1) = \int_0^\infty e^{-t} dt = 1 = 0!$

- $\Gamma(2) = 1 \cdot \Gamma(1) = 1 = 1!$
- $\Gamma(3) = 2 \cdot \Gamma(2) = 2 = 2!$
- $\Gamma(n) = (n-1)!$ for all positive integers n .

Half-integer values (stated):

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \quad \Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\sqrt{\pi}, \quad \Gamma\left(\frac{5}{2}\right) = \frac{3}{4}\sqrt{\pi}.$$

Connection to the Laplace transform. Substituting $u = st$ (so $t = u/s$, $dt = du/s$) in the definition of $\mathcal{L}\{t^n\}$:

$$\mathcal{L}\{t^n\} = \int_0^\infty e^{-st} t^n dt = \frac{1}{s^{n+1}} \int_0^\infty e^{-u} u^n du = \frac{\Gamma(n+1)}{s^{n+1}}, \quad n > -1.$$

This extends the integer formula $\mathcal{L}\{t^n\} = n!/s^{n+1}$ to all real $n > -1$.

5.1 Gamma Function Values Table

n	$\Gamma(n)$	$\mathcal{L}\{t^{n-1}\} = \Gamma(n)/s^n$
1	1	$1/s$
2	1	$1/s^2$
3	2	$2/s^3$
$1/2$	$\sqrt{\pi}$	$\sqrt{\pi}/s^{1/2}$
$3/2$	$\sqrt{\pi}/2$	$\sqrt{\pi}/(2s^{3/2})$
$5/2$	$3\sqrt{\pi}/4$	$3\sqrt{\pi}/(4s^{5/2})$

What Did We Learn?

The Gamma function is the natural extension of the factorial to real (and complex) numbers. Every time you need $\mathcal{L}\{t^n\}$ for a non-integer n , you use $\Gamma(n+1)/s^{n+1}$. The reduction formula $\Gamma(n+1) = n\Gamma(n)$ lets you reduce any half-integer argument to $\Gamma(1/2) = \sqrt{\pi}$.

6 Error Function (Brief Introduction)

The Error Function and Its Laplace Transform

Definition.

$$\operatorname{erf}(t) = \frac{2}{\sqrt{\pi}} \int_0^t e^{-u^2} du.$$

Key properties:

- $\operatorname{erf}(0) = 0$
- $\operatorname{erf}(\infty) = 1$

- $\operatorname{erf}(-t) = -\operatorname{erf}(t)$ (odd function)

Where it appears: heat conduction in semi-infinite rods (Fourier's law gives a solution in terms of erf), Gaussian probability distributions (normal CDF is expressed via erf), diffusion problems.

Laplace transform (result stated):

$$\mathcal{L}\left\{\operatorname{erf}\left(\sqrt{t}\right)\right\} = \frac{1}{s\sqrt{s+1}}.$$

This result requires a more advanced derivation involving the convolution theorem and will be revisited in Unit V.

7 Bessel Functions (Brief Introduction)

Bessel Function of the First Kind

Series definition:

$$J_0(t) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2} \left(\frac{t}{2}\right)^{2m} = 1 - \frac{t^2}{4} + \frac{t^4}{64} - \dots$$

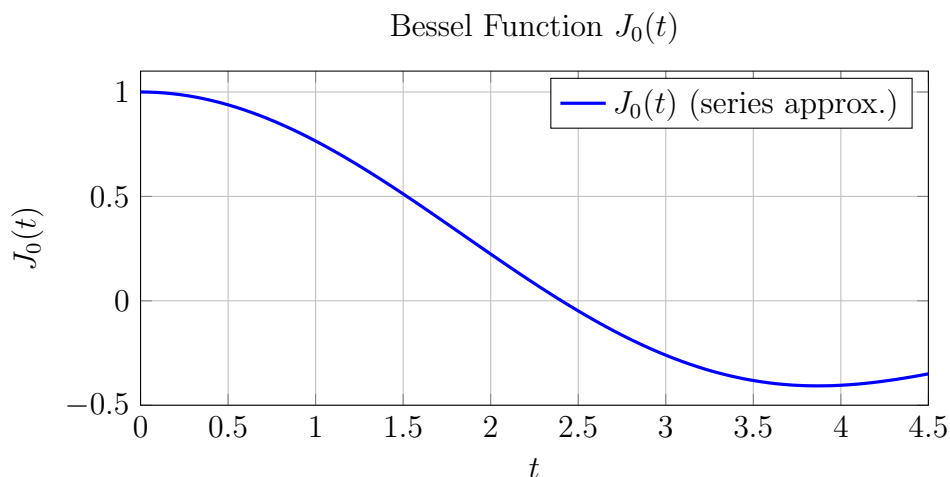
Behaviour: $J_0(t)$ oscillates with decreasing amplitude — like a damped cosine — arising in cylindrical geometries (vibrating drumheads, heat flow in cylinders, electromagnetic waveguides).

Laplace transform (result stated):

$$\mathcal{L}\{J_0(t)\} = \frac{1}{\sqrt{s^2 + 1}}.$$

This result arises in vibration and wave propagation problems and will be used in Unit V applications.

7.1 Plot of $J_0(t)$



What Did We Learn?

$J_0(t)$ is bounded (unlike exponentials) but oscillatory — it crosses zero infinitely many times. Its Laplace transform $1/\sqrt{s^2 + 1}$ reveals the resonant frequency structure embedded in the series definition.

8 Worked Examples

8.1 Example 1: Piecewise Function, Unit Steps, and the Laplace Transform

Example 1

Problem. Let

$$f(t) = \begin{cases} t^2 & 0 \leq t < 2 \\ 4 & t \geq 2 \end{cases}.$$

Write $f(t)$ using unit step functions, then find $\mathcal{L}\{f(t)\}$ using the Second Shifting Theorem.

Step 1 — Identify the pieces.

$g_1(t) = t^2$, $g_2(t) = 4$, switch at $a = 2$.

Step 2 — Apply the formula.

$$f(t) = g_1(t) + [g_2(t) - g_1(t)] u(t - 2) = t^2 + (4 - t^2) u(t - 2).$$

Step 3 — Rewrite $(4 - t^2)$ in terms of $(t - 2)$.

Let $\tau = t - 2$, so $t = \tau + 2$.

$$4 - t^2 = 4 - (\tau + 2)^2 = 4 - \tau^2 - 4\tau - 4 = -\tau^2 - 4\tau.$$

Therefore

$$4 - t^2 \Big|_{t=\tau+2} = -(t - 2)^2 - 4(t - 2).$$

So

$$f(t) = t^2 + [-(t - 2)^2 - 4(t - 2)] u(t - 2).$$

Step 4 — Apply $\mathcal{L}$ term by term.

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \mathcal{L}\{t^2\} - \mathcal{L}\{(t - 2)^2 u(t - 2)\} - 4 \mathcal{L}\{(t - 2) u(t - 2)\} \\ &= \frac{2}{s^3} - e^{-2s} \cdot \frac{2}{s^3} - 4 e^{-2s} \cdot \frac{1}{s^2} \\ &= \boxed{\frac{2}{s^3} - e^{-2s} \left(\frac{2}{s^3} + \frac{4}{s^2} \right)}. \end{aligned}$$

What Did We Learn?

Always expand $(g_2(t) - g_1(t))$ as a polynomial in $(t - a)$ before applying the Second Shifting Theorem. Trying to transform t^2 or t alongside $u(t - 2)$ directly violates

the form $g(t-a)u(t-a)$ required by the theorem.

8.2 Example 2: Combined Delta and Step Functions

Example 2

Problem. Find $\mathcal{L}\{3\delta(t-1) + 2u(t-2)\}$ step by step.

Step 1 — Linearity of the Laplace transform.

$$\mathcal{L}\{3\delta(t-1) + 2u(t-2)\} = 3\mathcal{L}\{\delta(t-1)\} + 2\mathcal{L}\{u(t-2)\}.$$

Step 2 — Apply individual transforms.

$$\mathcal{L}\{\delta(t-1)\} = e^{-1 \cdot s} = e^{-s} \text{ (with } a = 1 \geq 0\text{)}.$$

$$\mathcal{L}\{u(t-2)\} = \frac{e^{-2s}}{s} \text{ (with } a = 2\text{)}.$$

Step 3 — Combine.

$$\mathcal{L}\{3\delta(t-1) + 2u(t-2)\} = 3e^{-s} + \frac{2e^{-2s}}{s}.$$

What Did We Learn?

The Laplace transform of a sum of impulse and step terms is handled term by term via linearity. Note the structural difference: δ gives e^{-as} (no s denominator) while $u(t-a)$ gives e^{-as}/s (one power of s in the denominator, reflecting integration).

8.3 Example 3: Fractional Power via the Gamma Function

Example 3

Problem. Compute $\mathcal{L}\{t^{3/2}\}$ using the Gamma function. Express the answer in terms of π .

Step 1 — Apply the formula.

For $n = 3/2 > -1$:

$$\mathcal{L}\{t^{3/2}\} = \frac{\Gamma(3/2 + 1)}{s^{3/2+1}} = \frac{\Gamma(5/2)}{s^{5/2}}.$$

Step 2 — Evaluate $\Gamma(5/2)$.

Using the reduction formula twice:

$$\Gamma\left(\frac{5}{2}\right) = \frac{3}{2} \Gamma\left(\frac{3}{2}\right) = \frac{3}{2} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{3}{4} \sqrt{\pi}.$$

Step 3 — Write the final answer.

$$\mathcal{L}\{t^{3/2}\} = \frac{3\sqrt{\pi}/4}{s^{5/2}} = \boxed{\frac{3\sqrt{\pi}}{4 s^{5/2}}}.$$

What Did We Learn?

To find $\mathcal{L}\{t^n\}$ for any $n > -1$: write $\Gamma(n+1)/s^{n+1}$, then reduce $\Gamma(n+1)$ using $\Gamma(n+1) = n\Gamma(n)$ until you reach either $\Gamma(1) = 1$ or $\Gamma(1/2) = \sqrt{\pi}$.

9 Excel Example

Numerical Approximation of $\Gamma(3/2)$ Using the Trapezoidal Rule

Recall $\Gamma(3/2) = \int_0^\infty t^{1/2} e^{-t} dt$. The integrand decays very rapidly; integrating from 0 to 10 captures virtually all the area. Set up $t = 0.01, 0.11, 0.21, \dots, 9.91$ (step $\Delta t = 0.10$).

Column A: t	Column B: $f(t) = t^{0.5} \cdot e^{-t}$	Column C: Cumulative Trapezoid Sum
0.01	=A2^0.5*EXP(-A2)	=0
0.11	=A3^0.5*EXP(-A3)	=C2+(B2+B3)/2*(A3-A2)
0.21	=A4^0.5*EXP(-A4)	=C3+(B3+B4)/2*(A4-A3)
$\vdots$	$\vdots$	$\vdots$
10.01	=A102^0.5*EXP(-A102)	=C101+(B101+B102)/2*(A102-A101)

Comparison to exact value:

Method	Result	Error
Excel trapezoidal (100 steps, [0.01, 10.01])	≈ 0.8857	$\approx 0.06\%$
Exact: $\Gamma(3/2) = \sqrt{\pi}/2$	0.88623...	—

Excel Takeaway

The numerical integral of $t^{0.5}e^{-t}$ over $[0, 10]$ converges to $\sqrt{\pi}/2 \approx 0.8862$. This confirms that $\mathcal{L}\{t^{1/2}\}|_{s=1} = \Gamma(3/2)/1^{3/2} = \sqrt{\pi}/2$, i.e., the Gamma function value we use symbolically is numerically verifiable with a simple spreadsheet.

10 Python Example

The script below uses `scipy.special`, `sympy`, and `matplotlib` to explore every special function from this topic.

Listing 1: Special Functions in Laplace Transforms

```

1 import sympy as sp
2 from scipy.special import gamma as sc_gamma, j0, erf as sc_erf
3 import numpy as np
4 import matplotlib.pyplot as plt
5
6 # 1. Gamma function values

```

```

7 ns = [1, 2, 3, 0.5, 1.5, 2.5]
8 print("n      Gamma(n)      Approx")
9 print("-" * 35)
10 for n in ns:
11     val = sc.gamma(n)
12     print(f"n={n:<5}   {val:.6f}   {val:.4f}")
13 # Expected output:
14 # n=1      1.000000   1.0000
15 # n=2      1.000000   1.0000
16 # n=3      2.000000   2.0000
17 # n=0.5    1.772454   1.7725   (= sqrt(pi))
18 # n=1.5    0.886227   0.8862   (= sqrt(pi)/2)
19 # n=2.5    1.329340   1.3293   (= 3*sqrt(pi)/4)
20
21 #      2. Symbolic Laplace transform of t^(3/2) using sympy
22
23 t, s = sp.symbols('t s', positive=True)
24 f = t**sp.Rational(3, 2)
25 F = sp.laplace_transform(f, t, s)
26 print("\nL{t^(3/2)} =", F[0])
27 # Expected: 3*sqrt(pi)/(4*s**(5/2))
28
29 #      3. Plot J0(t)
30
31 t_vals = np.linspace(0, 20, 400)
32 j0_vals = j0(t_vals)
33
34 plt.figure(figsize=(8, 4))
35 plt.plot(t_vals, j0_vals, 'b-', linewidth=1.8, label=r'$J_0(t)$')
36 plt.axhline(0, color='k', linewidth=0.7)
37 plt.xlabel('t')
38 plt.ylabel(r'$J_0(t)$')
39 plt.title('Bessel Function of the First Kind, Order 0')
40 plt.grid(True)
41 plt.legend()
42 plt.tight_layout()
43 plt.savefig('j0_plot.png', dpi=150)
44 print("\nPlot saved to j0_plot.png")
45 # Expected: oscillating curve starting at 1, first zero near t
46 #           =2.40
47
48 #      4. erf(1)
49
50 print("\nerf(1) =", sc.erf(1.0))
51 # Expected: erf(1) = 0.8427007929497148
52
53 print("\nAll computations complete.")

```

Python Takeaway

The `scipy.special.gamma` function confirms the hand-computed Gamma values, including $\Gamma(1/2) = \sqrt{\pi} \approx 1.7725$. `sympy.laplace_transform` returns the exact symbolic form $3\sqrt{\pi}/(4s^{5/2})$, matching our pen-and-paper derivation. `erf(1) \approx 0.8427` confirms the error function saturates rapidly towards 1.

11 Viva-Style Oral Questions (8 Questions with Answers)

1. Define the unit step function $u(t - a)$.

Answer. $u(t - a) = 0$ for $t < a$ and $u(t - a) = 1$ for $t \geq a$. It represents an ideal switch that turns on at $t = a$.

2. What is $\mathcal{L}\{u(t - a)\}$?

Answer. $\mathcal{L}\{u(t - a)\} = e^{-as}/s$ for $s > 0$, derived by evaluating $\int_a^\infty e^{-st} dt = e^{-as}/s$.

3. State the sifting property of the Dirac delta function.

Answer. For any function f continuous at $t = a$: $\int_{-\infty}^\infty f(t) \delta(t - a) dt = f(a)$. The delta function “sifts out” the value of f at the single point a .

4. What is $\mathcal{L}\{\delta(t - a)\}$ and how is it derived?

Answer. $\mathcal{L}\{\delta(t - a)\} = e^{-as}$ ($a \geq 0$). It follows from the sifting property: $\int_0^\infty e^{-st} \delta(t - a) dt = e^{-sa}$.

5. What does the reduction formula $\Gamma(n + 1) = n\Gamma(n)$ give for positive integers?

Answer. Applying it repeatedly from $\Gamma(1) = 1$ gives $\Gamma(n) = (n - 1)!$ for any positive integer n . The Gamma function is thus the extension of the factorial to real numbers.

6. What is $\Gamma(1/2)$?

Answer. $\Gamma(1/2) = \sqrt{\pi}$. This is a standard result derived using the Gaussian integral $\int_0^\infty e^{-u^2} du = \sqrt{\pi}/2$.

7. How do you write the piecewise function $f(t) = t^2$ for $0 \leq t < 3$, $f(t) = 9$ for $t \geq 3$, using unit step functions?

Answer. $f(t) = t^2 + (9 - t^2)u(t - 3)$. Rewriting: $9 - t^2 = -(t - 3)^2 - 6(t - 3)$, so $f(t) = t^2 - [(t - 3)^2 + 6(t - 3)]u(t - 3)$.

8. Name one engineering application each for the unit step function, the Dirac delta, and the Gamma function.

Answer.

- *Unit step*: modelling a voltage source that switches on at $t = a$ in a circuit.
- *Dirac delta*: modelling a hammer blow (impulsive force) applied instantaneously at $t = a$.

- *Gamma function:* computing $\mathcal{L}\{t^{1/2}\}$ for a fractional-power signal in a filter analysis.

12 Descriptive Questions (5 Exam-Style Questions)

1. **Derive $\mathcal{L}\{u(t-a)\}$ from first principles.**

Starting from the definition, $\mathcal{L}\{u(t-a)\} = \int_0^\infty e^{-st} u(t-a) dt$. Since $u(t-a) = 0$ for $t < a$ and 1 for $t \geq a$, the integral reduces to $\int_a^\infty e^{-st} dt$. Evaluating: $\left[-\frac{e^{-st}}{s}\right]_a^\infty = 0 + \frac{e^{-as}}{s} = e^{-as}/s$, valid for $s > 0$.

2. **Derive $\mathcal{L}\{\delta(t-a)\}$ using the sifting property.**

By definition, $\mathcal{L}\{\delta(t-a)\} = \int_0^\infty e^{-st} \delta(t-a) dt$. Applying the sifting property with $f(t) = e^{-st}$ (continuous everywhere) and $a \geq 0$, the integral equals $e^{-s \cdot a} = e^{-as}$.

3. **Prove $\Gamma(n+1) = n \Gamma(n)$ by integration by parts.**

$\Gamma(n+1) = \int_0^\infty t^n e^{-t} dt$. Set $u = t^n$, $dv = e^{-t} dt$; then $du = nt^{n-1} dt$, $v = -e^{-t}$. IBP gives $[-t^n e^{-t}]_0^\infty + n \int_0^\infty t^{n-1} e^{-t} dt$. The boundary term is 0 for $n > 0$ (since $t^n e^{-t} \rightarrow 0$ as $t \rightarrow \infty$ and is 0 at $t = 0$). Hence $\Gamma(n+1) = n \Gamma(n)$.

4. **Write the piecewise function $f(t) = \begin{cases} \sin t & 0 \leq t < \pi \\ 0 & t \geq \pi \end{cases}$ using unit steps and find $\mathcal{L}\{f(t)\}$.**

$f(t) = \sin t - \sin t u(t-\pi)$. Now $\sin t = -\sin(t-\pi)$ (since $\sin(t-\pi) = -\sin t$), so $\sin t u(t-\pi) = -\sin(t-\pi) u(t-\pi)$. Hence $f(t) = \sin t + \sin(t-\pi) u(t-\pi)$. Transforming: $\mathcal{L}\{f(t)\} = \frac{1}{s^2+1} + e^{-\pi s} \cdot \frac{1}{s^2+1} = \frac{1+e^{-\pi s}}{s^2+1}$.

5. **Compute $\mathcal{L}\{t^{5/2}\}$ using the Gamma function.**

$\mathcal{L}\{t^{5/2}\} = \Gamma(5/2+1)/s^{5/2+1} = \Gamma(7/2)/s^{7/2}$. $\Gamma(7/2) = (5/2)\Gamma(5/2) = (5/2)(3/2)\Gamma(3/2) = (5/2)(3/2)(1/2)\sqrt{\pi} = 15\sqrt{\pi}/8$. Therefore $\mathcal{L}\{t^{5/2}\} = \frac{15\sqrt{\pi}}{8s^{7/2}}$.

13 Practice Problems (6 Problems with Answer Hints)

1. Find $\mathcal{L}\{u(t-3)\}$.

Hint: Use $\mathcal{L}\{u(t-a)\} = e^{-as}/s$ with $a = 3$. **Answer:** e^{-3s}/s .

2. Find $\mathcal{L}\{5\delta(t-2)\}$.

Hint: Linearity; $\mathcal{L}\{\delta(t-a)\} = e^{-as}$ with $a = 2$. **Answer:** $5e^{-2s}$.

3. Find $\mathcal{L}\{t^{-1/2}\}$ using the Gamma function.

Hint: $n = -1/2 > -1$ is valid. $\mathcal{L}\{t^{-1/2}\} = \Gamma(1/2)/s^{1/2} = \sqrt{\pi}/\sqrt{s}$. **Answer:** $\sqrt{\pi/s}$.

4. Write the three-piece function $f(t) = \begin{cases} 1 & 0 \leq t < 2 \\ t & 2 \leq t < 5 \\ 0 & t \geq 5 \end{cases}$ using unit step functions and

find $\mathcal{L}\{f(t)\}$.

Hint: $f(t) = 1 + (t-1)u(t-2) + (-t)u(t-5)$. Rewrite: $(t-1) = (t-2) + 1$ and $-t = -(t-5) - 5$; apply Second Shifting Theorem. **Answer:** $\frac{1}{s} + e^{-2s}\left(\frac{1}{s^2} + \frac{1}{s}\right) - e^{-5s}\left(\frac{1}{s^2} + \frac{5}{s}\right)$.

5. Find $\mathcal{L}\{(t-1)u(t-1) - (t-3)u(t-3)\}$.

Hint: Apply the Second Shifting Theorem term by term. **Answer:** $\frac{e^{-s} - e^{-3s}}{s^2}$.

6. Find $\mathcal{L}\{e^{-t}\delta(t-2)\}$.

Hint: Use the sifting property: $\int_0^\infty e^{-st} e^{-t} \delta(t-2) dt = e^{-2} e^{-2s}$. **Answer:** $e^{-2} e^{-2s} = e^{-2(s+1)}$.

14 MCQs (5 Questions)

1. What is $\mathcal{L}\{u(t-a)\}$?

- (a) $\frac{1}{s}$
- (b) $\frac{e^{as}}{s}$
- (c) $\frac{e^{-as}}{s}$
- (d) e^{-as}

Explanation: The integral $\int_a^\infty e^{-st} dt = e^{-as}/s$; the factor e^{-as} (not e^{as}) comes from the lower limit $a > 0$.

2. What is $\mathcal{L}\{\delta(t)\}$?

- (a) 0
- (b) $\frac{1}{s}$
- (c) e^{-s}
- (d) 1

Explanation: $\mathcal{L}\{\delta(t-a)\} = e^{-as}$; at $a = 0$ this gives $e^0 = 1$.

3. What is the value of $\Gamma(1/2)$?

- (a) 1
- (b) π
- (c) $\sqrt{\pi}$

(d) $1/2$

Explanation: $\Gamma(1/2) = \sqrt{\pi}$ is derived from the Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.

4. **The sifting property states that $\int_{-\infty}^{\infty} f(t) \delta(t - a) dt$ equals:**

(a) 0

(b) $f(0)$

(c) $\delta(a)$

(d) $f(a)$

Explanation: By definition, $\delta(t - a)$ is zero everywhere except $t = a$, so the integral extracts (“sifts”) the value $f(a)$.

5. **The piecewise function $f(t) = \begin{cases} 0 & t < 3 \\ 1 & t \geq 3 \end{cases}$ can be written as:**

(a) $\delta(t - 3)$

(b) $u(t) - u(t - 3)$

(c) $u(t - 3)$

(d) $u(t + 3)$

Explanation: $u(t - 3)$ is precisely 0 for $t < 3$ and 1 for $t \geq 3$, matching the given definition exactly.

15 Common Mistakes

Common Mistakes in Topic 07

Mistake	What Students Do	Correct Approach
Wrong transform of $u(t-a)$	Write $\mathcal{L}\{u(t-a)\} = 1/s$, forgetting the exponential factor	Always include e^{-as} : the result is e^{-as}/s .
Treating $\delta(t)$ as an ordinary function	Substitute a finite “value” for $\delta(t)$ or differentiate it like e^{-t}	$\delta(t)$ is a distribution; use the sifting property, never evaluate it pointwise.
Wrong Gamma-factorial relationship	Write $\Gamma(n) = n!$	The correct formula is $\Gamma(n) = (n-1)!$; equivalently $\Gamma(n+1) = n!$.
Forgetting the condition $a \geq 0$	Claim $\mathcal{L}\{\delta(t-a)\} = e^{-as}$ for negative a	The formula holds only for $a \geq 0$; for $a < 0$ the lower limit of the Laplace integral is 0, not $-\infty$.
Not rewriting in terms of $(t-a)$	Apply $\mathcal{L}\{g(t)u(t-a)\}$ directly as $e^{-as}G(s)$	First rewrite $g(t) = h(t-a)$ for some h ; then $\mathcal{L}\{h(t-a)u(t-a)\} = e^{-as}H(s)$.

16 Quick Recap

Quick Recap — Topic 07: Special Functions

- **Unit step:** $u(t-a) = 0$ for $t < a$, 1 for $t \geq a$; $\mathcal{L}\{u(t-a)\} = e^{-as}/s$; used to model switches and represent piecewise inputs compactly.
- **Dirac delta:** $\int \delta(t-a) dt = 1$, sifting property $\int f(t)\delta(t-a) dt = f(a)$; $\mathcal{L}\{\delta(t-a)\} = e^{-as}$; models instantaneous impulses.
- **Gamma function:** $\Gamma(n) = \int_0^\infty t^{n-1}e^{-t} dt$; reduction: $\Gamma(n+1) = n\Gamma(n)$; $\Gamma(1/2) = \sqrt{\pi}$, $\Gamma(3/2) = \sqrt{\pi}/2$; gives $\mathcal{L}\{t^n\} = \Gamma(n+1)/s^{n+1}$ for all $n > -1$.
- **Error function:** $\operatorname{erf}(t) = \frac{2}{\sqrt{\pi}} \int_0^t e^{-u^2} du$; $\mathcal{L}\{\operatorname{erf}(\sqrt{t})\} = 1/(s\sqrt{s+1})$; arises in heat conduction and probability.
- **Bessel function:** $J_0(t) = \sum_{m=0}^\infty \frac{(-1)^m}{(m!)^2} (t/2)^{2m}$; $\mathcal{L}\{J_0(t)\} = 1/\sqrt{s^2+1}$; arises in vibration and wave problems with cylindrical geometry.
- **Piecewise $\rightarrow$ unit steps:** $f(t) = g_1(t) + [g_2(t) - g_1(t)]u(t-a)$; always rewrite $g_2(t) - g_1(t)$ as a function of $(t-a)$ before transforming.
- **Key engineering roles:** step function $\rightarrow$ switched circuits; delta function $\rightarrow$ impulse response (Green’s function); Gamma function $\rightarrow$ fractional-power signals and filters.